

Ashwin Chembu

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PROFILE

Full-stack engineer and **applied AI builder** focused on reliable products, **RAG/search systems**, backend services, data pipelines, and clean user experiences for complex workflows.

TECHNICAL SKILLS

Front End: React, Next.js, TypeScript, JavaScript, Node.js, CSS, styled-components, HTML, Bootstrap

Back End: Python, Java, C, C++, MySQL, NoSQL, FastAPI, Flask, Express.js, MongoDB, Redis

AI/Data: SQL, Snowflake, PySpark, Azure Databricks, LangChain, OpenAI, AWS Bedrock, RAG, Power BI

Developer Tools: Git, Docker, AWS S3/Glue/Athena, Postman, Agile Methodology, Twilio, Leaflet, Figma/FigJam

EXPERIENCE

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2025 - Present

Built applied AI, data engineering, and MDM solutions for healthcare technology programs

Python | SQL | React | FastAPI | Snowflake | AWS | LangChain | Docker | Git

- **Built AI/RAG assistants** and backend services with **React, FastAPI, AWS data services, Redis**, and **document-ingestion workflows** for enterprise knowledge search.
- **Developed Snowflake SQL pipelines** for **master data extraction, identifier matching, specialty joins, address comparison, API connectivity**, and **quality-control reporting** across **327K+ records**.
- **Designed entity matching** and **license reconciliation** frameworks across **67K+ records** using **NPI, DEA, HIN, 340B, state-license**, and **fuzzy-search signals** for **stewardship review**.
- **Defined layered ETL/MDM architecture** across **landing, staging, transform, canonical**, and **publish patterns**, including **DQ checks, delta logic**, and **history tracking**.
- **Produced client-ready artifacts** including requirements, source-to-target mappings, executive decks, decision logs, and process documentation for **MDM and CRM workflows**.

Founder & Lead Developer | Tallyrus

Remote | Jun 2023 - Present

Built NLP, RAG, and full-stack tooling for automated essay feedback

Python | LangChain | Pinecone | OpenAI | MongoDB | Flask | React | Node.js

- **Developed Tallyrus** using **Python, NLP, LangChain, Pinecone**, and **OpenAI** to automate essay grading with **retrieval-augmented workflows**.
- **Built scalable backend services** with **Flask, MongoDB**, and **S3**, optimizing data management, security, and response times by **30%**.
- **Designed React interfaces**, rubric workflows, and feedback loops that helped educators review essay quality faster while preserving consistent scoring logic.
- **Led product planning, user testing**, iterative development, and fundraising, raising **five figures** to support product growth.

Data Analyst / Web Development Lead | Tech4Good @ UCSC

Santa Cruz, CA | Mar 2023 - Present

Built research tooling, analytics workflows, and collaboration features for social impact

Data Modeling | Python | R | SQL | React | Node.js

- **Led development** of a **FigJam Chat Widget** and **CSV Importer Plugin** using **JavaScript, React, Node.js, Python**, and **Pandas** for nearly **4,000 users**.
- **Conducted data analysis** with **Python, Pandas, NumPy, R, SQL**, and **Power BI** for research workflows and interactive visualization.
- **Built plugin and widget features** for real-time collaboration, structured imports, and clean user flows across research and classroom environments.
- **Improved collaborative tooling** and data-processing workflows, contributing to **40% higher engagement** and **25% better processing efficiency**.

PROJECTS

Lume - AI Meeting Intelligence Platform

San Francisco, CA | Dec 2025 - Present

- **Built a full-stack AI meeting intelligence platform** using **React 18, Node.js/Express, MongoDB, AWS S3**, and **WebSockets** to capture, store, and query meetings.
- **Architected a RAG-based Q&A engine** with **vector embeddings, intent routing, evidence extraction**, and **multi-pass generator/critic refinement**.
- **Designed dynamic Project Brain knowledge bases** to learn **terminology, relationships**, and context from transcripts and seed files.
- **Implemented transcript-aware search, document-grounded answers**, and reusable project context to improve **retrieval precision** across meetings and files.

Echo

Cambridge, MA / Davis, CA | Sep 2023 - Mar 2024

- **Collaborated with a professor-led research team** to build a **digital verification platform** for online identity authentication and secure media workflows.
- **Integrated facial detection ML APIs, AWS S3/Multer storage**, and backend services for secure image handling.
- **Developed backend architecture** using **MongoDB, Express, Flask**, and **Python SDK integrations**, with a **React/Node** frontend.
- **Connected product flows** to a research blockchain stack for identity proof creation, persistence, and frontend verification states.

File System

Davis, CA | Sep 2023 - Oct 2023

- **Developed ECS150-FS**, a **File Allocation Table** file system with virtual disk emulation.
- **Implemented file-system operations** including **mounting, unmounting, creating, deleting, descriptors**, and file handling.
- **Managed metadata and file data** to preserve consistent file-system state and deepen **C programming** fundamentals.
- **Tested edge cases for allocation, persistence**, and file operations to strengthen low-level systems debugging and memory discipline.

EDUCATION

University of California - Davis | **B.S. Computer Science** | **GPA: 3.7**

Davis, CA | Graduated Jun 2025

Relevant Foundation: Algorithms, Databases, Distributed Systems, Operating Systems, AI/ML, Data Structures

CERTIFICATIONS

AWS Certified Cloud Practitioner | **Google Data Analytics Certificate** | Databricks Data Engineering | Apache Spark Programming

PUBLICATION

ICER - Evaluating Exploratory Reading Groups for Supporting Undergraduate Research Pipelines in Computing - Jun 12, 2024

LINKS

Website: ashchembu.com | **LinkedIn:** linkedin.com/in/ashwin-chembu-a5292b205 | **GitHub:** github.com/ashwinchembu

ACTIVITIES

Leadership: Tau Beta Pi Treasurer | **Honors:** Dean's Honor